ElderConnect - A Smart Friend for Health, Safety & Harmony

END TERM VII SEMESTER SYNOPSIS REPORT

Submitted in partial fulfillment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

The NorthCap University

by

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Session 2025-2026

CERTIFICATE

This is to certify that the Project Synopsis entitled, **ElderConnect - A Smart Friend for Health, Safety & Harmony** submitted by Harsha(22CSU081), Himanshi Yadav(22CSU085), Leepakshi Bhatia(22CSU107), Prabhdeep Singh(22CSU133) to **The NorthCap University, Gurugram, India,** is a record of bonafide synopsis work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of the University.

Dr. Swati Gupta

Assistant Professor(Sr. Scale), CSE

Date: 29-11-2025

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Abstract

ElderConnect is a cutting-edge artificial intelligence-enabled virtual companion created to tackle many of the complex needs of older adults around issues of health, safety, socialization, and sustainable living. As we are living longer on a global scale, many older adults are experiencing chronic conditions, increased susceptibility to accidents, and social isolation that compounds the impact of these issues on caregivers. Current solutions usually address one kind of function (to be used singularly), such as medication reminders, fall detection, and location tracking. There is a lack of a comprehensive, integrated platform.

ElderConnect aggregates these key services in a single application that is easy to use. The eldercare solution offers health support through medication management, monitoring vitals, and individualized health and wellness support.. ElderConnect comprises safety through fall detection, geo-fencing, and emergency responses. ElderConnect promotes mental wellness through AI driven companionship, reminiscence therapy, and spontaneous video calls with a family member. Finally, ElderConnect has a sustainability model by promoting energy efficiency, making use of solar power, and promoting an eco-friendly lifestyle.

ElderConnect brings together all of these functionalities, allowing for older adults to retain their independence while having continuous connection with caregivers and loved ones. For caregivers, ElderConnect provides comprehensive real-time monitoring and alerts for action and thereby continues to lessen the burden of caring for older adults.

Introduction

a. Background

An increasingly older population faces complicated challenges that transcend medical concerns to include risks to safety, social isolation, and a decline in cognitive engagement. While families and caregivers often try their best to fill these gaps, their ability to do so effectively can be limited by the constraints of time, caregiver burden, and a less-than-ideal support structure. Current digital solutions are often limited to one specific function (to detect a fall, remind a patient to take medication, or track locations), and no solution has surfaced that is comprehensive, integrated, user-friendly, and senior-centric. ElderConnect aims to help solve these challenges through an integrated ecosystem of new product features designed for older adults. The platform provides comprehensive features and functionality beyond preventive health reminders, including health tracking in real time via Bluetooth-enabled health device (e.g. blood pressure, oxygen, thermometer), AI meal and medication logging, and caregiver dashboard -alerts, and eco-metrics features. Safety features include SOS panic buttons, fall detection via IoT sensors, geo-fencing, and GPS travel monitoring. Emotional, social features are an important aspect of ElderConnect, which includes the capability for family video chatting, sharing of media, reminiscence therapy and peer event matching. Cognitive and recreation function and game and puzzles, guided meditation and relaxation exercises are also features of ElderConnect. To maximize emotional and social well-being, the platform facilitates video connection to family, sharing of multimedia content, reminiscence therapy activities, and peer group connections. Mental stimulation[2] and relaxation through leisure is also provided with options such as games, puzzles, relaxation instructions and meditation practices. The product also enhances accessibility and entertainment options with audiobooks, podcasts, large text options and hearing aid compatibility. An environmental focus has been added through eco-mode, solar charging capabilities and a daily tip feature encouraging sustainable living behaviour.

b. Feasibility Study

The viability of the ElderConnect platform is based on being technologically feasible, user-friendly, and highly demanded in the market. From the technology perspective, the system leverages state-of-the-art development tools and architectures: React Native/Expo for spawning easy-to-use mobile applications, Firebase for secure user authentication, a backend on Nest.js and Express to handle requests, data persistence with PostgreSQL for scalable, reliable storage, IoT devices for vital signs measurement health monitoring features, falls detection with TensorFlow.js, and location services including geo-fencing by Google Maps APIs for emergency response capabilities.While media is saved and provided from AWS S3 and Firebase, social interaction uses WebRTC and Twilio technologies for video communication features. Lastly, AI-enabled features are used in personalized health tracking to record medications and meals. In selecting its design approach, ElderConnect acknowledges the critical significance that accessibility plays for the

elderly. This will further be enhanced with features such as voice-command input, gesture recognition, font size increase, magnifier, flashlight, and all with simple single-tap navigation for users. Environmental sustainability also features as part of the overall design of ElderConnect, including solar charging, energy-saving eco modes, and carbon footprint measuring tools-all responsible and highly innovative in this platform.

From a commercial standpoint, the increasing number of older adults seeking eldercare solutions, coupled with falling costs of IoT hardware and open source AI, makes ElderConnect scalable and provable in real world environments. Uniquely, ElderConnect provides a complete functionality suite, dashboards for care-focused users, and a more integrated approach to dated systems, making it easy to upload ElderConnect into real world eldercare contexts and situations.

Study of Existing Solutions

There are many technology-based eldercare solutions today, each focused on an aspect of elderly wellness. An example is Medisafe [1], an app that is now commonly used to help users manage their daily medications by sending reminders and alerts, tracking a schedule, and providing real time medication adherence data to other caregivers.

While Medisafe is effective in its role of improving medication adherence, that is primarily what Medisafe is focused on—medication scheduling—rather than being a health and safety solution in a more general sense.. Life360 [1] was developed mainly for family communication and safety, providing tracking of location over time, the ability to contact family during a car accident, and other notifications to connected family members.

Life360 is focused on family safety and is missing health monitoring and emotional engagement which makes it less useful if it is considered a stand-alone eldercare product. BSafe focuses on personal safety with SOS alerts, the ability to share live video, and a "follow-me" function to let users situate themselves and broadcast their status to others during emergencies. Medisafe and Life360 provide partial eldercare support but lack integration [1], [5]. Bsafe [1] does not provide health monitoring or some of the social interactions elderly persons require for wellness. Most eldercare technologies, such as CarePredict, Lively, and Philips Lifeline, are designed to detect falls, monitor movement, and rapidly respond to emergencies through subscription services. These devices have limited broader social engagement features.

General-purpose devices like the Apple Watch provide additional functionality through heart rate monitoring and integrations with health apps, targeting active seniors, yet are generally more expensive and require some level of digital literacy. CarePredict is differentiated by its AI-powered behavioral analytics, real-time location tracking, monitoring of staff performance, and predictive wellness insights in support of both older adults and senior community staff with more comprehensive and proactive features than classic alert-only systems. Philips Lifeline supports emergency response along with a digital caregiving network to connect family and professional caregivers-adding to the care circle experience. Lively provides reliable personal medical alert services but remains focused primarily on safety features.

Comparison with Existing Software Solutions

Feature	Existing Solutions	ElderConnect
Health Monitoring	Apple Health, Medisafe	Integrated (medication, vitals, nutrition)
Safety Features	bSafe, Life360	SOS, fall detection
Emotional Engagement	Limited	AI companionship, therapy, video calls
Integration & Accessibility	Fragmented, high digital literacy required	Unified, designed for low digital literacy

Gap Analysis

Although there are a wide variety of eldercare technologies available as mobile apps, wearable devices, and monitoring services, technology still has large gaps that could improve its ability to help older adults and their caregivers. Studies show accessibility and sustainability remain ignored in eldercare solutions [2], [6].

Limited Creation of Service Options

Most current eldercare products focus primarily on a particular function or service, requiring the user to manage several different apps or devices that are disconnected to address their needs. This can cause unnecessary bother and confusion for often older or more frail consumers. The ideal technology would combine health care, safety, emotional well being, and lifestyle management into a single product that is easy to learn and use.

Access Limitations to Technology

Most eldercare technologies assume that older adults can reasonably work with a digital product. There is a constant gap in the usability of features, including small text and limited support for vision and hearing impairments and very few have accessible features in place. Minimal options, including voice commands and gesture navigation, large fonts, or include audiological devices for hearing. There are certainly good options that appeal to older adults but are not ubiquitous, ultimately compromising the accessibility of these products to patients or caregivers who may experience physical or cognitive challenges.

Neglect of Environmental responsibility

While safety and health are priorities of users of elder care technology, the concern for environmentally sustainability is very limited. Emerging eldercare technologies do not consider energy-efficient operation, incorporation of solar-powered charging, or have sharing mechanics that encourage environmentally responsible use. It is possible to consider responsible innovation when there is an increasing focus on sustainability globally.

Dependency on Internet Connectivity

Many eldercare applications require constant internet access, which can be unreliable or unavailable in rural or low-connectivity areas. The lack of offline functionality for critical features such as emergency alerts or medication reminders poses serious risks, since uninterrupted digital services cannot always be guaranteed for seniors.

Problem Statement

As they age, older persons face a variety of difficulties, including deteriorating health, mobility issues, safety concerns, and social isolation [2]. Their freedom and quality of life are negatively impacted by these difficulties. This circumstance can occasionally put caretakers under excessive pressure. While many current technologies provide crucial features like location monitoring, medication reminders, and fall detection, the majority of these technologies are fragmented and perform discrete tasks. Because of the nature of the technology, older persons must traverse a multitude of devices or apps, which can be difficult and perplexing for them, especially if they have poor digital abilities or sensory deficiencies.

Additionally, most existing systems are not designed for accessibility or environmental responsibility. According to NITI Aayog [3], the demand for senior care solutions is rapidly increasing in India. Interfaces are rarely designed with older adults in mind, there are very limited options for offline emergency support, and few, if any, tools promote sustainability. Caregivers routinely experience additional challenges related to monitoring and do not have an integrated dashboard or means to track the well-being of older adults in real-time. This clarifies the need for a one-stop shop solution that easily integrates health monitoring, safety, emotional support, and sustainability.

ElderConnect is designed to provide an integrated solution that meets a variety of older adult and caregiver needs, a solution that allows older adults to live better lives and caregivers to have less burdens.

Objectives

The primary goal of ElderConnect is to provide older adults with a friendly and supportive virtual companion that would allow them to experience better management of their health on a daily basis, feel safe, and emotionally secure. The site also considers the elderly and includes easy screens, easy reminders, and other subdued instructions that acknowledge the age related difficulties like low vision, poor memory or mobility.

The second objective is to make the seniors feel safe at home or elsewhere. Such features as early fall detection, prompt SOS response, sharing their location during outings, and real-time updates to the caregivers provide them with more confidence and control. This also contributes towards minimization of unnecessary hospital referral and postponement of long term institutional care.

The third objective is to build social connections and become less lonely. ElderConnect is a smartphone-based solution that is focused on alleviating the burden of caregivers by using automated assistance and smart monitoring. It also fosters a healthy lifestyle, high energy lifestyle and enables the elders to keep in touch with the family by use of communication tools and friendly interaction features.

Lastly, the platform has been designed in a cost-effective, scalable, and adaptable manner within communities, languages, and care settings. It can collaborate with local services and available health systems to aim at aiding in planning of the public health, cutting down on the expenses of long term care, and above all, sustaining the dignity and autonomy of the elderly.

Tools/platform Used

ElderConnect revolves around a modern scalable and multi layer technology stack, which is used to provide health monitoring, safety alerts and social interactions functionality. The tools and platforms used included the following:

Frontend / Mobile Application

- React Native / Expo, to create an easy-to-use and cross-platform mobile application user interface.
- UI/UX tools, designed to develop simplified screens that are easy to use by the elderly (large fonts, understandable navigation).

Application Programming Languages

- Nest.js & Express, to process API requests and process backend logic.
- PostgreSQL, for secure and scalable user account, health data, reminder and location logs storage.

AI/ML & Data Processing

- TensorFlow.js, applied to fall-detection algorithm and behavioural analysis.
- AI-based meal tracking and medication documentation, uses AI models to help users (elderly) track their health.

Design Methodology

The design and research process used to develop ElderConnect is done in such a way that it forms an integrated, accessible and elder-focused platform of eldercare.

The steps followed are:

- **Problem Identification** :
  The health decline, safety risks, mobility issues, loneliness, and access to digital resources were analyzed in detail as the challenges encountered by the elderly people.

- **Study of Existing Solutions and Literature Review:**
  The apps such as Medisafe, Life360, bSafe, Apple Health and CarePredict and Philips Lifeline that already exist were considered.
  This showed that the majority of solutions are dedicated to a single functional area, are not accessible and are not combined with emotional well-being or sustainability.

- **Gap Analysis:**
  Significant gaps that have been identified are:
  - Disjointed technology that needs several applications.
  - Low usability (poor accessibility i.e., small text, no voice support).
  - Poor emotional and social engagement instruments.

- **Requirement Gathering & Feasibility Study:**

  Technical feasibility was considered using:

- React Native/ Expo in accessibility.
- Firebase, PostgreSQL for scalable architecture, Nest.js.
- TensorFlow.js for Fall detector.
- Twilio/WebRTC communication.
- Vitals monitoring from IoT and Bluetooth gadgets.
- Voice recognition, oversized text, magnifier, single-tap navigation.
- Offline safety support
- The increasing demand of integrated eldercare solutions was proved by commercial feasibility.

- **System Design:**
  There was an interface that was planned to have a unified interface.
  Developed health tracking, medication, appointments, fall detection, SOS and family

communication modules.

- **Development Phase:**
  Installed front end screens and a navigation.
  Compiled backup APIs of reminders, vitals, and safety notifications.
  AI models formed to detect falls and intelligent health monitors were developed.

- **Testing & Evaluation:**
  Tried with older people in terms of usability (large fonts, accessibility features).
  Stable video calling is ensured.
  Assessed performance of low-end devices.

- **Deployment & Improvement:**
  Prepared app for deployment
  Collected user feedback
  Future enhancements (additional sensors, further analytics) that are planned.

Outcomes

ElderConnect provides a nurturing and trustworthy supportive environment to the elderly by uniting the accessible technology in an accessible, significant manner. The platform will take into account the actual, daily needs of older adults with a combination of safety functions, social networking, and simple monitoring of the health condition to simplify and make life more comfortable.

The presence of a friendly person to converse with and mechanisms of maintaining a social life can significantly alleviate loneliness and anxiety among elderly people. This can be overcome by providing conversational assistance and methods to maintain contact with loved ones provided through ElderConnect.

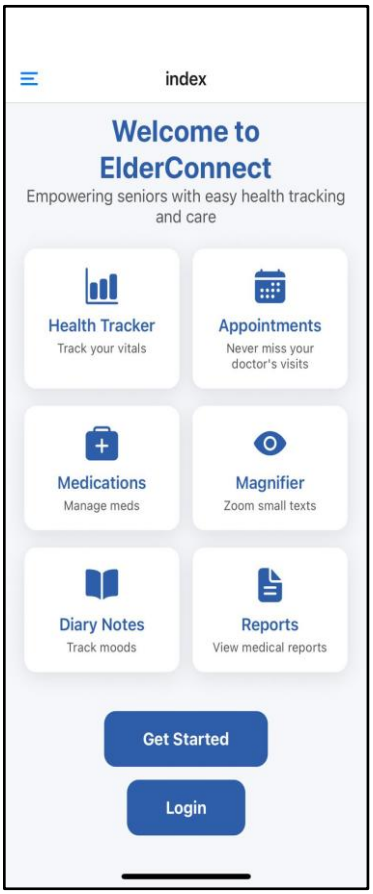
Safety is another key focus. Additional benefits such as fall detectors, SOS notifications and GPS positioning make the seniors feel more secure when staying at home or going on a walk. These types of tools also enable them to live longer and minimize the risks of serious falls.

Simple reminders and vitals tracking allow seniors to cope with chronic illnesses to manage their everyday health. This will eliminate complications and minimise chances of unwarranted visits to the hospital.

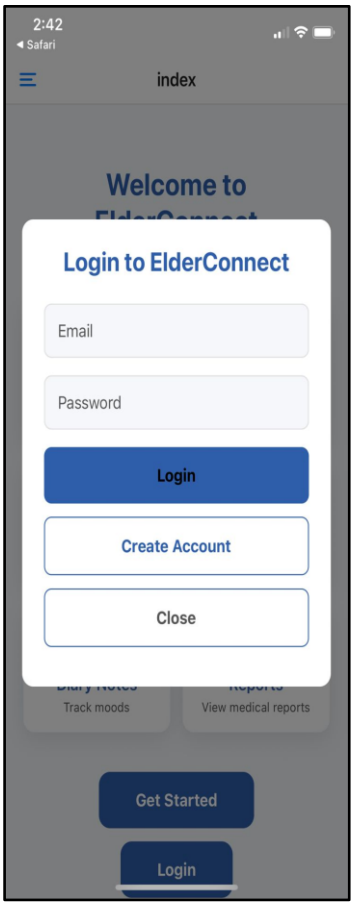
Besides assisting the seniors themselves, ElderConnect assists families and caregivers. Caregivers are less stressed and feel more at ease that something many studies on digital health have documented.

The platform will also be able to lower the long-term healthcare expenses by providing early warning of changes in their health, enhancing safety, and promoting healthy practices. And since it is accessible in nature, ElderConnect helps close the digital divide, making technology more accessible and accessible to the seniors.

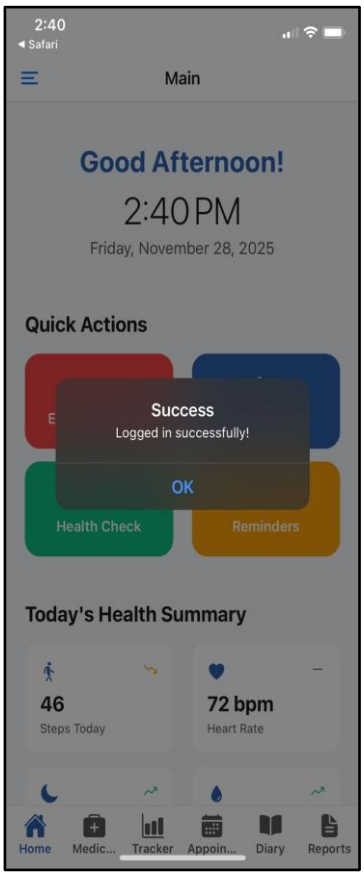
Github Repository - https://github.com/prabhdeep114198/ElderConnect



(i) Home Page

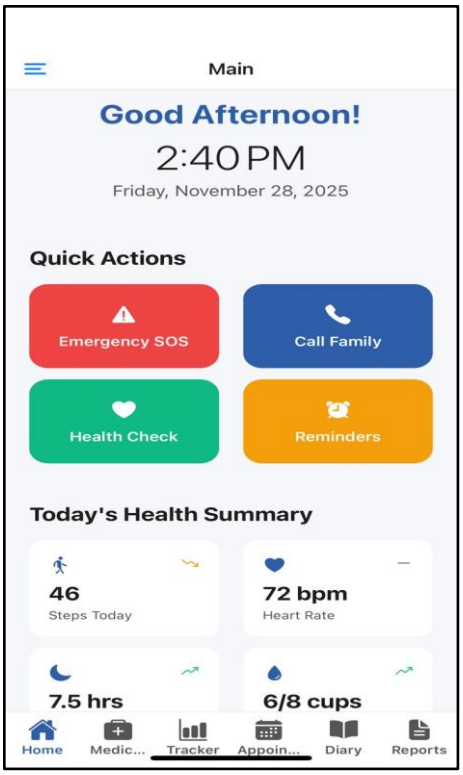


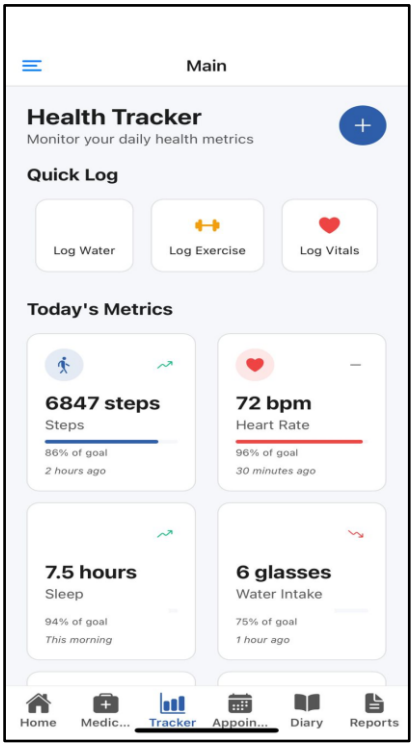
(ii) Login Page



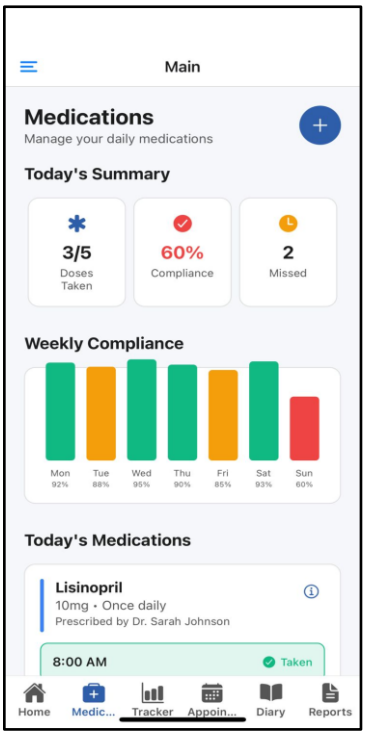
(iii) Successful Login

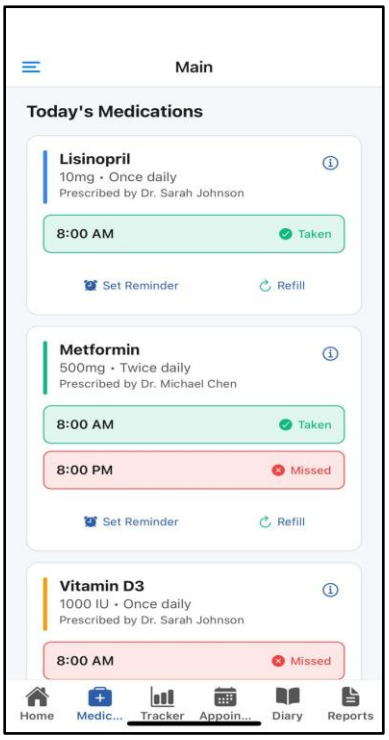
To access the app, Users can either create an account or add registered details.

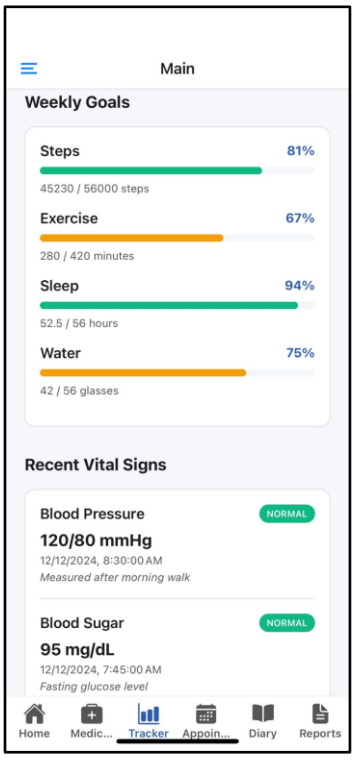


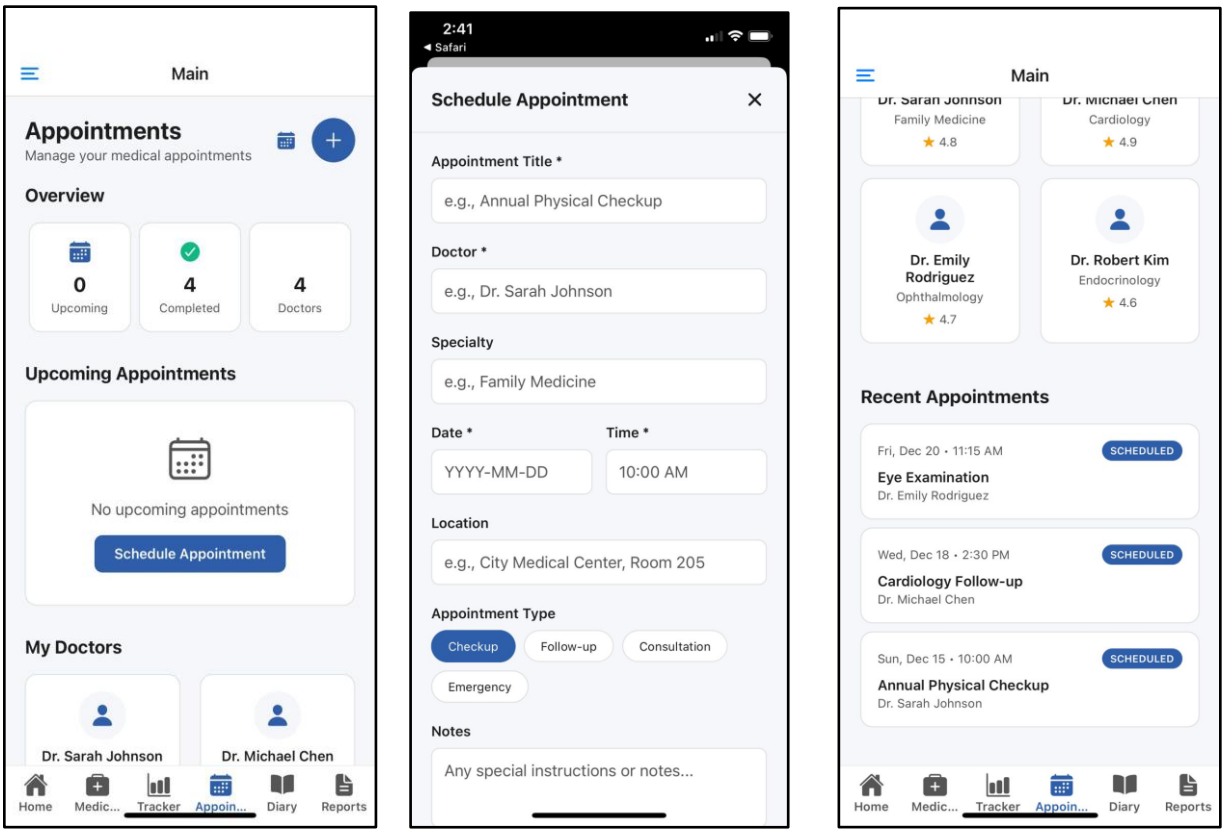


Users can contact their family, track their health and add reminders for medicines.

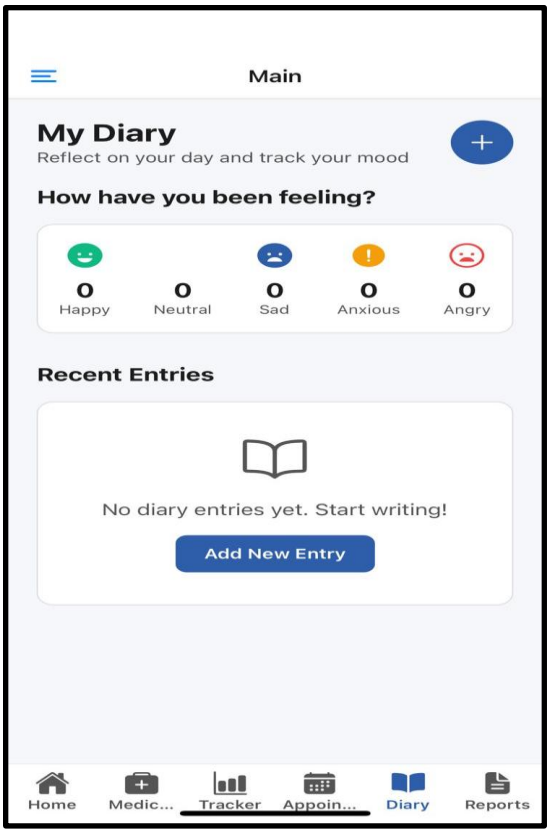


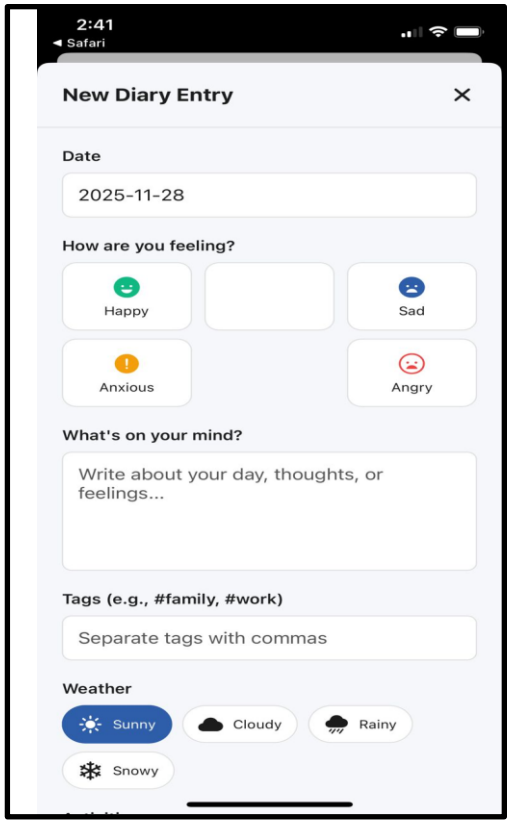




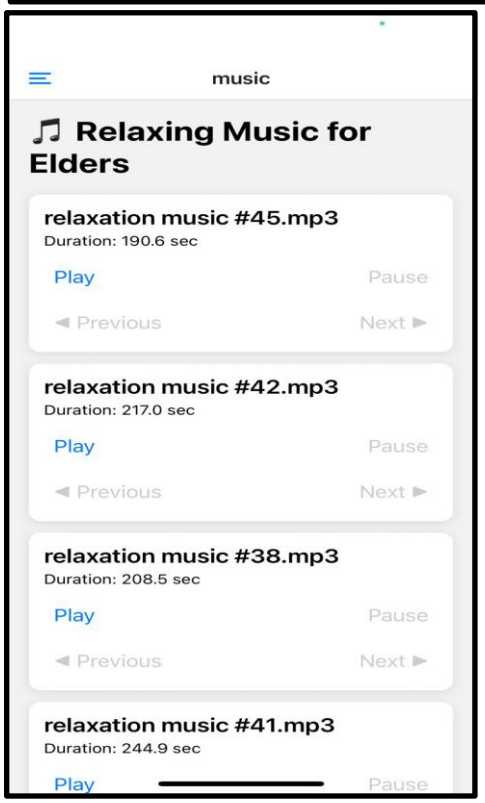


Users can add or check their appointments easily.


Main
My Diary
Reflect on your day and track your mood
How have you been feeling?
0 Happy
0 Neutral
0 Sad
0 Anxious
0 Angry
Recent Entries
No diary entries yet. Start writing!
Add New Entry
Home
Medic...
Tracker
Appoin...
Diary
Reports


2:41
Safari
New Diary Entry
Date
2025-11-28
How are you feeling?
Happy
Sad
Anxious
Angry
What's on your mind?
Write about your day, thoughts, or feelings...
Tags (e.g., #family, #work)
Separate tags with commas
Weather
Sunny
Cloudy
Rainy
Snowy

Events
Social Gatherings Near You
ACETECH Delhi
4/12/2025 – 7/12/2025
expos
Sourced from predicthq.com - ACETECH is India's leading trade fair for architecture, design, and building materials.
Christmas Eve
24/12/2025 – 25/12/2025
public-holidays
Sourced from predicthq.com - Christmas Eve is the day before Christmas Day and falls on December 24 in the Gregorian calendar.
Shivaji Jayanti
19/2/2026 – 20/2/2026
public-holidays
Sourced from predicthq.com
Vasant Panchami
23/1/2026 – 24/1/2026
public-holidays
Sourced from predicthq.com - Vasant Panchami, or India's spring festival, is a Hindu event that includes special rituals and highlights the start of spring.
Valentine's Day
14/2/2026 – 15/2/2026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🎵 Relaxing Music for Elders
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Pause
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Next
relaxation music #42.mp3
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Play
Pause
Previous
Next
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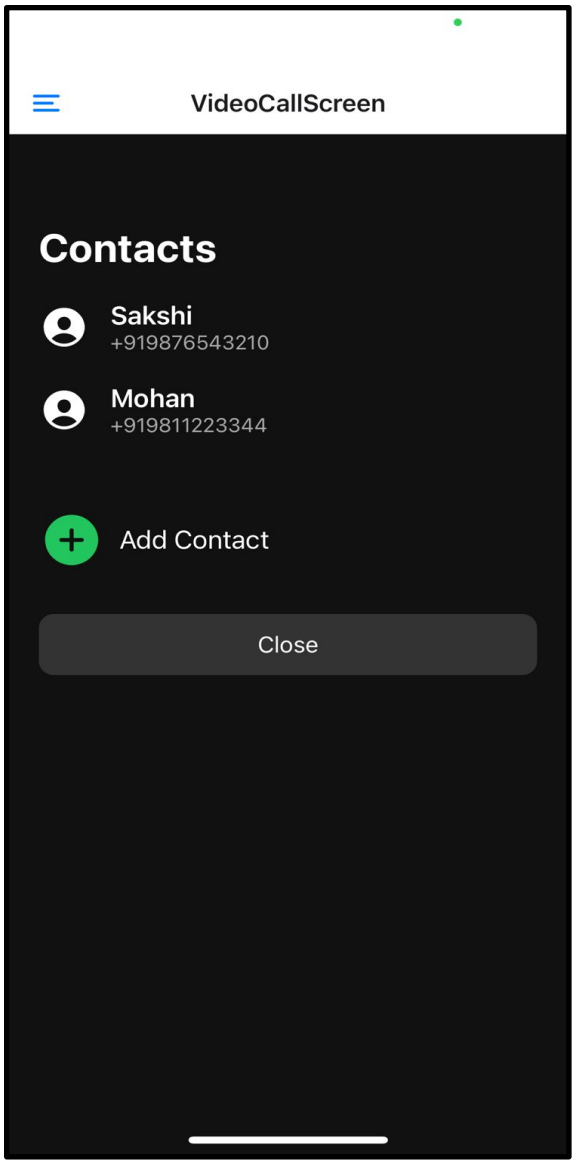
Duration: 208.5 sec
Play
Pause
Previous
Next
relaxation music #41.mp3
Duration: 244.9 sec
Play
Pause


VideoCallScreen
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Sakshi
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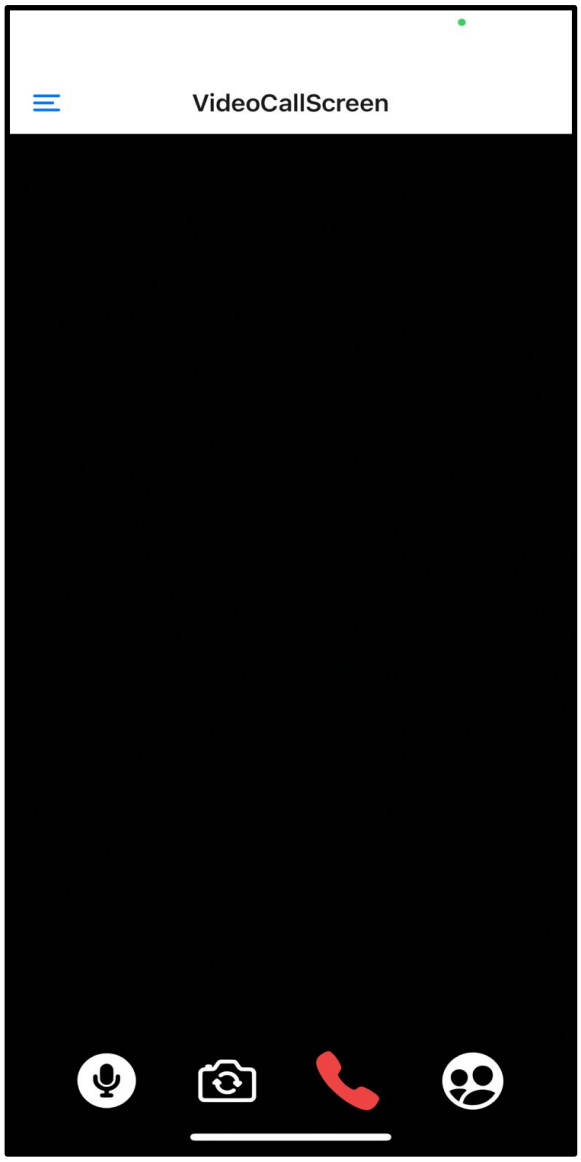
Mohan
+919811223344
Add Contact
Close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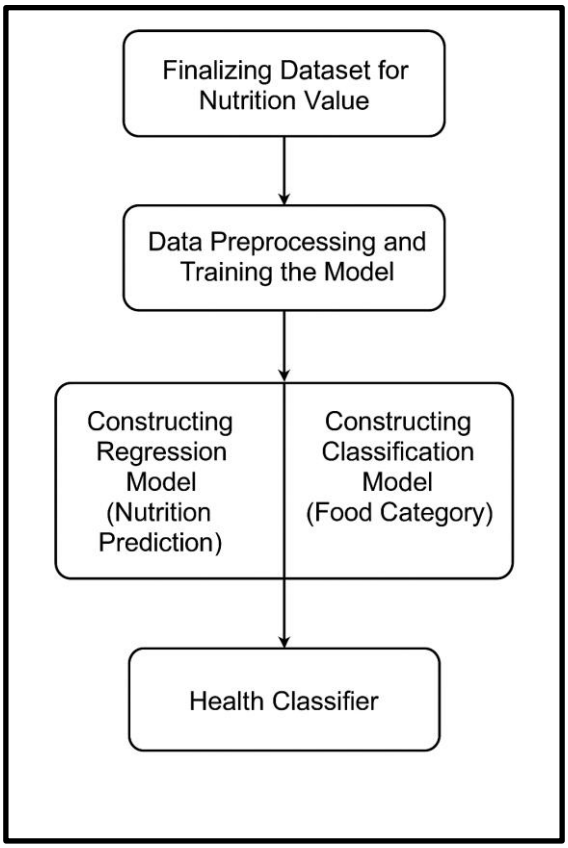
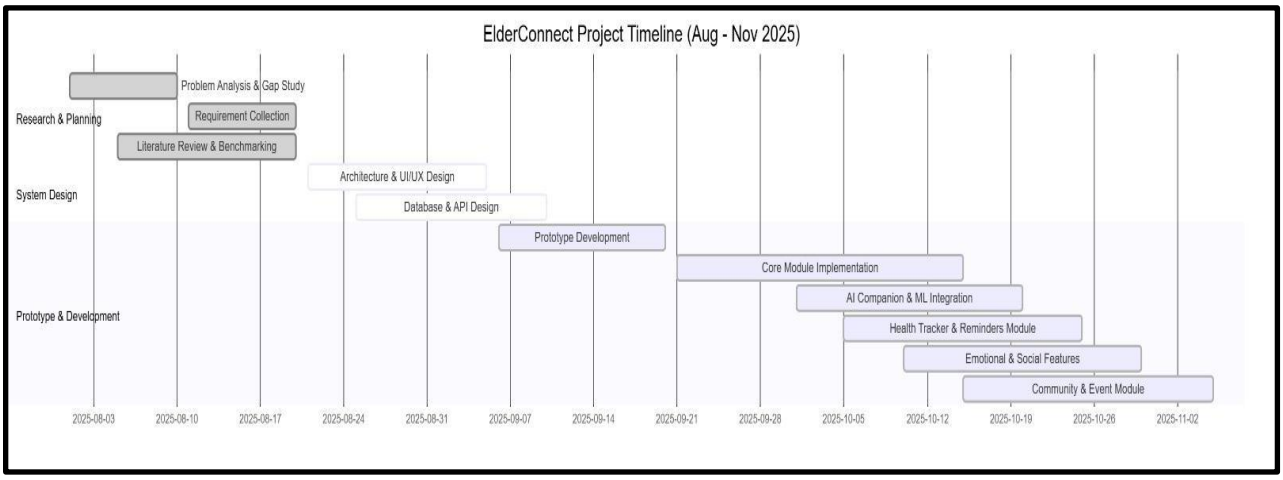

Finalizing Dataset for Nutrition Value
Data Preprocessing and Training the Model
Constructing Regression Model (Nutrition Prediction)
Constructing Classification Model (Food Category)
Health Classifier

Gantt Chart


ElderConnect Project Timeline (Aug - Nov 2025)
Research & Planning
System Design
Prototype & Development
Problem Analysis & Gap Study
Requirement Collection
Literature Review & Benchmarking
Architecture & UI/UX Design
Database & API Design
Prototype Development
Core Module Implementation
AI Companion & ML Integration
Health Tracker & Reminders Module
Emotional & Social Features
Community & Event Module
2025-08-03
2025-08-10
2025-08-17
2025-08-24
2025-08-31
2025-09-07
2025-09-14
2025-09-21
2025-09-28
2025-10-05
2025-10-12
2025-10-19
2025-10-26
2025-11-02

Responsibility Chart

Team Member	Responsibility
Harsha Saini (22CSU081)	Frontend Development & UI/UX
Himanshi Yadav (22CSU085)	AI and Data Analysis
Leepakshi Bhatia (22CSU107)	AI/ML Model Development & Testing
Prabhdeep Singh (22CSU133)	Frontend & Backend Development

References

- https://www.agespace.org/top-10-useful-apps-help-elderly
- https://www.oliveeldercare.com/what-are-the-problems-faced-by-elderly-people-in-our-society/
- https://www.niti.gov.in/sites/default/files/2024-02/Senior%20Care%20Reforms%20in%20India%20FINAL%20FOR%20WEBSITE_compressed.pdf
- https://www.jll.com/en-in/insights/elevating-the-golden-years-a-report-on-indias-senior-living-market-landscape
- https://yourstory.com/2024/12/reimagining-elder-care-startups-innovating-greying-population-senior
- https://www.agewellfoundation.org/?page_id=5768

Annexure I:

turnitin Page 2 of 30 - Integrity Overview Submission ID trn:oid:::21301:123093389

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

- **8** Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
0% Publications
6% Submitted works (Student Papers)

Annexure II:

Supervisor's comments

Annexure - 1

Week no.	Supervisor's Comment (about general project progress and individual contribution towards the project)	Supervisor's Signature with date
WEEK 1	Discussion about project Title & Detailed analysis (Mtt-Survey)	Swati 20/8/25
WEEK 2	Research and discussed about • Life 360 and My Fitness Pal Features • Medisafe and magnifying glass Feat.	Swati 27/8/25
WEEK 3	Research and discussed about audio book, podcast, relaxation integration with panic button and voice interface.	Swati 04/09/25
WEEK 4	Reviewed some research papers to understand the level of existing work in the domain & created flowchart.	Swati 11/9/25
WEEK 5	Finalized the feature flow, created the design and started working on the prototype.	Swati 16/9/25
WEEK 6	Worked on main features like health monitoring dashboard, daily reminders, Diary Notes.	Swati 23/9/25
WEEK 7	Implemented navigation flow between Home, Health, Safety and Social modules.	Swati 30/9/25
WEEK 8	Started backend structure planning for reminders, daily activity tracking and monitoring dashboards. Prepared API's for integration of health data.	Swati 3/10/25

WEEK 9	Implemented reminder scheduling module for medicines, doctor visits. Tested push notification functionality.	Swati 9/10/25
WEEK 10	Worked on Medications, Reports and Login modules and implemented their backend. Then worked on Magnifier and Video Call Screen.	Swati 15/10/25
WEEK 11	Added the events and music section for social interaction through free API's.	Swati 30/10/25
WEEK 12	Added the flashlight module or feature to the magnifier screen to make it easier for elders to read small text in low - light conditions.	Swati 7/11/25
WEEK 13	Searched, collected and reviewed datasets required for Diabetes Prediction, Fall detection, nutrition value & chatbot model development. Analyzed & finalized datasets.	Swati 15/11/25
WEEK 14	Developed and trained Diabetes Prediction Model. Performed data cleaning, feature selection, model training, accuracy evaluation and comparison of algo. to finalize best performing model.	Swati 21/11/25
WEEK 15	Created fall detection model by implementing sensor-based data processing & ML classification. Constructed regression model, classification model & health classifier for Nutrition Prediction & food category.	Swati 29/11/25
WEEK 16		
WEEK 17		